

Abstract

Flourishing in Diversity: The Promise and Limits of Agent-Based Models

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Can a handful of artificial agents following a few simple rules teach us anything about how real communities flourish? This dissertation answers by putting *agent-based models*—simulations in which simple agents interact, learn, and produce collective patterns—to work on three such questions: how an epistemic community can identify its experts, when a scientific community gains from sustained dissent, and whether a world of suffering could be beneficial for moral development. The members of these communities differ in expertise, in opinion, and in well-being, and in each case the question is how those differences can be turned into epistemic or moral goods.

The first two applications belong to social epistemology. Against Alice Huang’s model of expertise, I show that her skeptical verdict on track records depends on records that are over-trusted, immediate, and private; once trust is earned gradually, recognition delayed, or evidence made public, track records regain their value. Reassessing the Zollman effect—the finding that less communication can make a community more accurate—I argue that any policy lesson requires separate judgments about which success metric matters, the time frame, the size of the intervention, the difficulty of the problem, and the evidence available. The third application carries the method into the philosophy of religion: I build a novel agent-based model of the soul-making theodicy and show that, if freely helping the vulnerable builds virtue, then an intermediate amount of suffering, a low rate of severe evil, and a lawlike rather than targeted distribution of evil may each be connected

to soul-making goods in ways that aren't obvious at first glance.

Across the three, a cautionary lesson recurs. Agent-based models can make hidden assumptions explicit, reveal surprising mechanisms, test an argument's robustness, and prove that a possibility is coherent; but on their own they're rarely sufficient to settle a complex question of policy or metaphysics. Taken together, the applications show how such models can illuminate epistemic and moral flourishing—and where that illumination gives out.